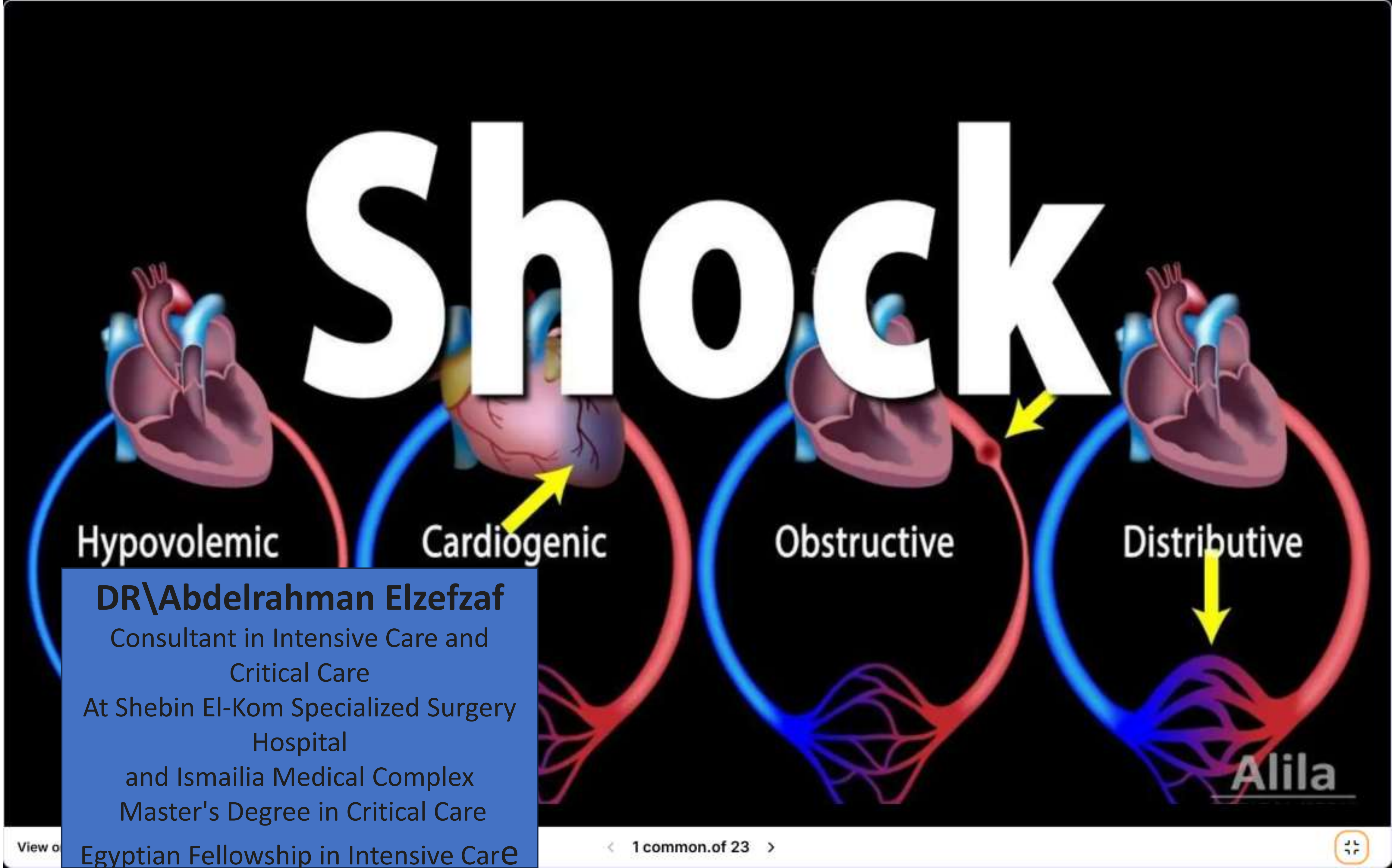

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Shock

- Shock is a life-threatening condition where there is inadequate tissue perfusion and oxygen delivery to meet cellular metabolic needs, leading to cellular and organ dysfunction.

Types of Shock

1) Hypovolemic Shock
2) Cardiogenic Shock
3) Distributive Shock
4) Obstructive Shock

1) Hypovolemic Shock

- Cause: Loss of blood or plasma volume (e.g., hemorrhage, dehydration, burns).
- Compensatory mechanisms: tachycardia, vasoconstriction, activation of RAAS and ADH.

Pathophysiology of Hypovolemic Shock

Decreased intravascular volume

↓

preload (↓)

↓

stroke volume (↓)

↓

cardiac output (↓)

↓

tissue perfusion (↓)

2) Cardiogenic Shock

- Cause: Cardiac pump failure (e.g., myocardial infarction, arrhythmias, cardiomyopathy).
- Elevated preload and afterload due to compensatory vasoconstriction worsen the cardiac workload.

Pathophysiology of Cardiogenic Shock

Impaired myocardial contractility

↓

stroke volume (↓)

↓

cardiac output (↓)

↓

perfusion despite adequate volume (↓)

3) Distributive Shock

It includes:

a. Septic
b. Anaphylactic
c. Neurogenic shock

Pathophysiology of Distributive Shock

Widespread vasodilation

↓

systemic vascular resistance (SVR) (↓)

↓

blood pressure and perfusion (↓) .

- Often associated with increased capillary permeability and fluid leakage.

Pathophysiology of SEPTIC Shock

Septic Shock:

Triggered by severe infection

↓

systemic inflammatory response

↓

vasodilation, capillary leak, and myocardial depression.

Pathophysiology of ANAPHYLACTIC Shock

Anaphylactic Shock:

Allergic reaction

↓

histamine release

↓

vasodilation and increased permeability.

Pathophysiology of NEUROGENIC Shock

Neurogenic Shock:

Spinal cord injury

↓

loss of sympathetic tone

↓

unopposed vagal activity

↓

vasodilation and bradycardia

4) Obstructive Shock

- Cause: Physical obstruction to blood flow (e.g., pulmonary embolism, cardiac tamponade, tension pneumothorax).
- Despite normal myocardial function, the obstruction prevents effective circulation.

Pathophysiology of Obstructive Shock

Impaired venous return or cardiac filling/output

↓

stroke volume (↓)

↓

cardiac output (↓)

↓

tissue perfusion (↓).

Stages of Shock

1. Initial Stage
2. Compensatory Stage
3. Progressive Stage
4. Irreversible Stage

1. Initial stage

1. Initial Stage:

Oxygen delivery (↓)

↓

shift to anaerobic metabolism

↓

lactic acidosis

- Subtle or no clinical signs

2. Compensatory stage

Activation of sympathetic nervous system (SNS)

↓

Tachycardia, vasoconstriction, increased respiratory rate

- Blood is shunted to vital organs (brain, heart).

3. Progressive stage

Persistent hypoperfusion

↓

worsening acidosis

- End-organ dysfunction begins (e.g., oliguria, confusion)

4. Irreversible stage

Severe cellular and organ damage

↓

Multi-organ failure

- Death inevitable without intervention

Management of Shock

General Approach (All Types):

- Airway, Breathing,
- Circulation (ABCs)
- Oxygen therapy
- IV fluid resuscitation
- Monitor vitals, urine output, lactate levels

Specific Treatments

1. Hypovolemic shock:

Rapid fluid replacement (crystalloids/blood) and control bleeding

2. Cardiogenic shock:

Inotropes (e.g., dobutamine), diuretics, revascularization (if MI), manage arrhythmias

3. Distributive shock:

Septic: Broad-spectrum antibiotics, vasopressors (e.g., norepinephrine)

Anaphylactic: Epinephrine IM, antihistamines, corticosteroids

Neurogenic: Fluids, vasopressors, atropine (if bradycardia)Obstructive: Relieve obstruction (e.g., thrombolysis, pericardiocentesis, chest tube)

4. Obstructive shock:

Relieve obstruction (e.g., thrombolysis, pericardiocentesis, chest tube)

Vital signs (normal values):

1-**GCS** 15(Eye 4,verbal 5, Motor 6)
2-**RBS** (fasting 70:120 , pp <180)
3-**Blood pressure** (around 120/80), invasvive or non invasive
4-**Heart Rate** (HR, pulse) (60:100) {ECG}
5-**Temperature** (Around 37c)
6-**Respiratory rate** (RR) (14:20) eMV, tracheostostomy
7- **O2 saturation** >96% , (nasal , mask ,venturi, high flow ,nasal cannula , cpap ,mv)
8-**Capnogram** (35:45)
9-**CVP** (6:8 mmHg)
10- **Urine output**
Fluid Balance =Intake(oral , ryle , gasrerostomy)-output (vomiting , stol , urine , drains)
Others : Nutrion status (oral , ryle , gasterostomy , PTPN , TPN)
Pass stool